# Jillanne Youssef

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## EDUCATION

**University of Waterloo, BASc in Mechanical Engineering** Expected April 2029

## SKILLS

**Mechanical Design:** CAD (NX, CATIA V6, SolidWorks), DFMA (Injection Molding, Die-Casting, CNC-Machining)
**Analysis:** Tolerance Analysis, FEA (Ansys Mechanical), CFD (Ansys Fluent), RCFA/FMEA, Mold Flow Simulation
**Manufacturing & Prototyping:** Machining (Lathe, Mill), 3D Printing (SLA, SLS, FDM), Waterjet & Laser Cutting
**Electrical & Software:** PCB Design (KiCad, Altium), Embedded (ESP32, Arduino, Raspberry Pi) C++, Python, MATLAB

## EXPERIENCE

**Google** May 2026 – Aug 2026 | Mountain View, California
*Product Design Engineering Intern - Pixel Hardware*

- Owned the design and validation of the Pixel Watch screen geometry for precision-molded glass, using prototype testing and supplier DFM to eliminate optical distortion while meeting wall-thickness and draft constraints.
- Saved $15K in vendor testing by developing a production-representative NMT test coupon and fixture that recreated a Pixel Watch titanium-resin interface, enabling seal-failure root-cause testing without complete watch builds.
- Developed an installed-condition drop test to validate dynamic FEA of a preloaded glass-filled plastic watch baffle, avoiding destruction of 15 low-volume prototypes per test cycle and saving $20K+ in prototype costs.
- Engineered a universal strain-test jig for resin dogbone specimens, using a mathematically derived elliptical profile and tool-free clamping to standardize 2% strain testing across 4 global suppliers and eliminate supplier-specific tooling.

**Level Home** Jan 2026 – Apr 2026 | Redwood City, California
*Product Design Engineering Intern*

- Reduced unit cost by $12 by replacing two accelerometers and FPCAs with integrated magnetometers and insert-molded magnets for lock retraction detection, yielding $12M projected savings.
- Improved capacitive touch reliability during EVT by designing an automated solenoid test fixture and PCBA, collaborating with EE and FW to resolve 14 issues across 2M+ validation cycles, resulting in 100% functional success.
- Resolved bolt binding observed in cycle testing by defining datums to the bolt travel path and tightening critical tolerances, reducing actuation force variability 25% and improving mechanism pass rate 15% prior to production.
- Led DFM for 20+ plastic and metal injection molded and die-cast parts; relocated parting lines and gates off cosmetic surfaces through supplier tooling reviews, eliminating cosmetic defects and tooling cost 8% while preserving ID intent.

**Tesla** Sep 2025 – Dec 2025 | Fremont, California
*Mechanical Design Engineering Intern - Lighting, Switches, Sensors*

- Led the design of a new program interior cabin light, using CATIA V6 for solid and surface modeling of Class A surfaces for injection molding, optimized for cost, ease of assembly, and user experience.
- Designed a 2-layer mixed-signal PCB from scratch in KiCad, specified and selected all components, hand-soldered and assembled the board, and optimized the design for low cost, DFM, and fabrication.
- Proposed and built an RGBW controller for internal company use, replacing off-the-shelf controllers by integrating a PCB, SLA-printed enclosure, and ESP32, streamlining design studio process by increasing LEDs per controller by 6x.

**Tesla** Jan 2025 – Apr 2025 | Palo Alto, California
*Mechanical Design Engineering Intern - Cell Equipment*

- Designed a high-temp magnetic clamping assembly for cell lamination, achieving 0.1 mm clearance through tolerance stackups and vibration-resistant design, boosting yield by 15% and cutting setup time from 30 to 5 seconds.
- Led mechanical, electrical, and controls design of a modular foil stretching machine with load-cell-based tension control, reducing foil wrinkling and creasing failures by 50%.
- Designed and built an electrical control box integrating Arduino, load cell, amplifier, and signal conditioner for closed-loop pneumatic control of nip rollers, maintaining roller nip force within ±2% of target under dynamic loading.